Lam Fung Academy · LF Academy

Primary 5 · 8th Lesson · Student Handout

Adding fractions with different denominators

Unit 3 · Fraction operation · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5A Unit 3 + Modern Education 5A Unit 10

Core Trap: T9 Fraction Operations - After the common division, the numerator forgets to expand simultaneously · The result is not reduced

SSPA Association: 🔴 High frequency The score test is compulsory every year, accounting for about 15-20% of Paper 1

Prerequisite knowledge: Class 7 (Comparison of different denominators and common fractions · LCM calculation · Equivalent fractions and extended fractions)

Objectives of this class:❶ Understand that different parts cannot be added directly ❷ Be proficient in the three-step method ❸ Addition of mixed numbers ❹ Application problems

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions（total 5 question · 5 minutes）

#	Question	Difficulty	Working Space（Show full working）
1	Calculate LCM(4, 6) = ?	Basic	
2	Calculate LCM(3, 5) = ?	Basic	
3	Divide $\frac{1}{2}$ and $\frac{1}{3}$ (write the two equivalent fractions after the common denominator)	Advanced	
4	Calculate $\frac{3}{8}+\frac{2}{8}=$? (Adding with the same denominator, P4 basics)	Basic	
5	Compare $\frac{2}{3}$ and $\frac{3}{5}$ which one is bigger? Write down the judgment process.	Advanced	

II.Core Knowledge ＋ Worked Examples

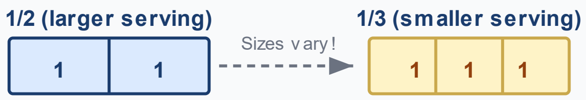
Knowledge point 1: Why can’t fractions with different denominators be added directly? 🔴SSPA

① The denominator represents "the number of parts the whole is cut into." Different denominators = different sizes of each portion

② For example: $\frac{1}{2}$ is cut into **2 parts** · $\frac{1}{3}$ is cut into **3 parts**——The two parts are different sizes!

③ To add, it must first become **the same parts**→ This is the **common fraction**

④ The essence of the common fraction: use "equivalent fractions" (extended fractions) to change the denominator into LCM



❌ Cannot directly 1/2 + 1/3 = 2/5

✅ Must be divided first: 1/2=3/6, 1/3=2/6 → 5/6

Cut the whole into 6 parts (LCM=6)

Example of trap detonation (the most important demonstration in this class)

calculate: $\frac{5}{6}-\frac{1}{4}=$?

❌ **Common mistakes (80% students)**

$$\frac{4}{2} = 2$$

Direct numerator minus numerator, denominator minus denominator

✅ **Correct solution**

$$\frac{7}{12}$$

LCM(6,4)=12 → $\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$

🧠 **Tip: "First understand the LCM, then multiply the numerator and keep the denominator unchanged. The answer must be simplified."**

⚠ **The most frequent error: adding and subtracting different denominators directly without common denominators. Remember: the denominators are different → be sure to connect the denominators first!**

⚠ **The second most frequent error: only changing the denominator when dividing the whole number, forgetting to expand the numerator simultaneously. "The molecules follow and ride"!**

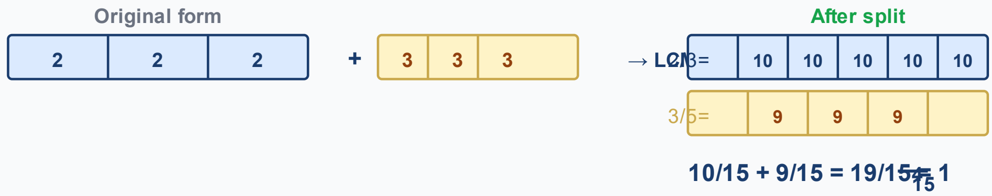
Knowledge Point - Worked Examples (write out ①LCM ②find a common denominator ③calculate ④simplify)

#	Question	Difficulty	Working Space
Example 1	$\frac{1}{4} + \frac{1}{2} = ?$	🌱	
Example 2	$\frac{1}{3} + \frac{1}{6} = ?$	🌱	

Knowledge point 2: General addition of fractions with different denominators (LCM is greater than the denominator itself) 🔴 SSPA required test

Standard common three-step method:

① **LCM |||SEP|||: Short division to find the least common multiple** ②Common branch**|||SEP|||: simplest fraction/false** → **band**: Simultaneous expansion of molecules ③**calculate**: Addition of numerators with the denominator unchanged ④**reduce**: Simplest fraction/false→band



LCM(3,5)=15 → 2/3=10/15, 3/5=9/15
10/15+9/15=19/15=1 and 4/15

①	②	③	④
Looking for LCM	**common points**	**calculate**	**reduce**
Find the lowest common multiple using short division	The numerator and denominator change simultaneously to LCM	Adding numerators with the denominator unchanged	Simplest false fraction→mixed fraction

example

Example 3: $\frac{2}{3} + \frac{3}{5} = ?$

example

Example 4: $\frac{3}{8} + \frac{5}{6} = ?$

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
6	$\frac{1}{2} + \frac{1}{3} = ?$	🌿	
7	$\frac{2}{3} + \frac{1}{4} = ?$	🌿	
8	$\frac{3}{5} + \frac{1}{2} = ?$	🌿	

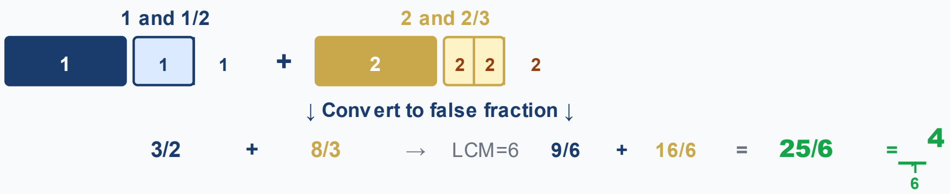
9	$\frac{3}{4} + \frac{1}{6} = ?$		
10	$\frac{5}{8} + \frac{1}{6} = ?$		

Knowledge Point 3: Addition of Different Denominators with Mixed Fractions

🔴 SSPA Advanced

Recommended method (conversion to improper fraction method - the least error-prone):

① Mixed fraction → Improper fraction (integer × denominator + numerator) ② Common fraction (find LCM) ③ Calculation ④ Answer → Mixed fraction + reduction



1 and 1/2 + 2 and 2/3 → 3/2 + 8/3 → 9/6 + 16/6 = 25/6 = 4 and 1/6

example

Example 5: $1\frac{1}{2} + 2\frac{2}{3} = ?$

example

Example 6: $2\frac{1}{3} + 1\frac{1}{2} = ?$

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
11	$1\frac{1}{4} + 2\frac{2}{3} = ?$		
12	$1\frac{2}{5} + 2\frac{3}{10} = ?$		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
13	$\frac{1}{4} + \frac{1}{8} = ?$	🌱	
14	$\frac{1}{3} + \frac{1}{9} = ?$	🌱	
15	$\frac{1}{2} + \frac{1}{4} = ?$	🌱	
16	$\frac{1}{5} + \frac{3}{10} = ?$	🌱	
17	$\frac{2}{5} + \frac{1}{3} = ?$	🌿	

Advanced layer (total 5 questions, 🚶🚀 choose do)

#	Question	Difficulty	Working Space
18	$\frac{3}{5} + \frac{1}{2} = ?$	🌿	
19	$\frac{5}{8} + \frac{1}{6} = ?$	🌿	
20	$\frac{3}{4} + \frac{1}{6} = ?$	🌿	
21	$1\frac{1}{4} + 2\frac{2}{3} = ?$	🌳	
22	$2\frac{1}{3} + 1\frac{1}{2} = ?$	🌳	

🌳 challenge layer (total 3 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
23	$2\frac{1}{3} + 1\frac{1}{2} + \frac{3}{4}$ =? (Mixture of three fractions, including mixed numbers)	🌳	

24	$\frac{1}{2} + \frac{2}{3} + \frac{1}{6} = ?$	🌳	
25	A rope is divided into three sections: the first section is $\frac{1}{3}$ meters, the second section is $\frac{1}{4}$ meters longer than the first section, and the third section is $\frac{5}{6}$ meters. How many meters is the total length of the rope? (You need to find the length of the second paragraph first)	⛰️	

Knowledge point 4: Application questions on addition of fractions (subject to sub-test word questions) 🔴 SSPA compulsory exam

Four steps to solve the problem:① **Circle keywords**("Total" "Total" = addition) ②**Column** ③ **Calculation of common score** ④ **Write a complete answer sentence**(Step points will be deducted if you do not write an answer sentence!)

example

Example 7: Xiao Ming has $\frac{1}{2}$ Pizzas, and Xiaohua has $\frac{1}{3}$ Pizzas. How many pizzas do they have in total?

applicationquestionpractice (all must do, column → find a common denominator → calculate → answer sentence)

#	Question	Difficulty	Working Space
26	Xiao Mei drank $\frac{1}{4}$ liters of orange juice, and Xiao Ming drank $\frac{1}{2}$ liters. How many liters did they drink in total?	🌿	
27	A box of chocolate, my sister ate $\frac{2}{5}$ boxes, my brother ate $\frac{1}{3}$ boxes. How many boxes did the two of them eat in total?	🌿	
28	The original water bottle contains $\frac{3}{4}$ liters, then pour $\frac{1}{3}$ liters into it. How many liters are there now? (Answer with mixed fractions)	🌳	

advancedapplicationquestion (🧗🚀 choose do, SSPAPaper 2commonquestion type)

#	Question	Difficulty	Working Space

29	The cake shop sold $\frac{2}{5}$ pieces in the morning and $\frac{1}{3}$ pieces in the afternoon. How many were sold in total throughout the day?	🌿	
30	The first section of rope is $1\frac{1}{4}$ meters, the second section is $2\frac{1}{3}$ meters. How many meters are the total length of the two sections? (Answer with fractions)	🌳	
31	|||SEP||| in the garden grows roses, $\frac{1}{4}$ grows chrysanthemums. What percentage of the garden is occupied by flowers? $\frac{2}{5}$ Plant chrysanthemums. What percentage of the garden is occupied by flowers?	🌳	
32	|||SEP||| in the granary is for white rice, $\frac{1}{6}$ is for brown rice, and $\frac{2}{5}$ is for red rice. What fraction of the granary is occupied by rice? What percentage is the remainder? $\frac{1}{3}$ Put red rice. What fraction of the granary is occupied by rice? What percentage is the remainder?	🌳	

IV. ⛰ Ultimate challenge area

#	Question	Difficulty	Working Space
⛰ 1	$\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{6}$ =? (Adding four fractions with different denominators - find LCM(2,3,4,6))	⛰	
⛰ 2	A bottle of juice contains $1\frac{1}{2}$ liter. I drank $\frac{1}{3}$ liters for breakfast and $\frac{1}{4}$ liters for lunch. How many liters are left? (This class is learning addition, but this question uses subtraction!)	⛰	
⛰ 3	Normal points account for |||SEP|||, tests account for |||SEP|||, exams account for |||SEP|||. What percentage of the total score do the three items account for? What's left? (Total score = 1) $\frac{2}{5}$, the test accounts for $\frac{1}{3}$, the exam accounts for $\frac{1}{4}$. What percentage of the total score do the three items account for? What's left? (Total score = 1)	⛰	

V. Class afterhomework

Basic must-do questions (total 6 questions, must write find a common denominator step)

#	Question	Difficulty	Working Space
H1	$\frac{1}{4} + \frac{1}{8} = ?$	🌱	
H2	$\frac{1}{3} + \frac{1}{9} = ?$	🌱	
H3	$\frac{1}{2} + \frac{1}{6} = ?$	🌱	
H4	$\frac{2}{3} + \frac{1}{4} = ?$	🌿	
H5	$\frac{3}{4} + \frac{1}{6} = ?$	🌿	
H6	Xiao Ming has $\frac{1}{3}$ pieces of chocolate, and Xiaohua gives him $\frac{1}{2}$ pieces. How many lines does Xiao Ming have now?	🌿	

Advanced choose doquestion (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H7	A bottle of juice $\frac{3}{4}$ liters, then pour $\frac{1}{3}$ liters of water. How many liters are there now? (Answer with fractions)	🌳	
H8	Compare $\frac{5}{6} + \frac{1}{4}$ and $\frac{1}{2} + \frac{2}{3}$ which one is bigger? How much difference?	🌳	
H9	$1\frac{2}{3} + 2\frac{1}{2} + \frac{3}{4} = ?$	⛰️	

VI. The Lessoncorecommon errorsummary

✅ Self-examination in this hall (tick after completion)

- ☐ I know the pitfalls of solving each knowledge point
- ☐ I can complete 🌱basic questions independently
- ☐ I can challenge 🌿advanced questions
- ☐ I remember the formula

🎯 Review of Learning Objectives - After completing this lesson you should be able to:

- ☐ Identify all trap types in our hall
- ☐ Solve 🌱basic questions independently (100% correct)
- ☐ Challenge 🌿Advanced questions (80%+ correct)
- ☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	**Add the different denominators directly**: $\frac{1}{2}+\frac{1}{3}=\frac{2}{5}$	Different denominators → Be sure to connect the denominators first!
2	**After splitting, the molecules forget to expand simultaneously.**	"Multiply the numerator" - multiply the denominator by the same number and multiply the numerator by the same number
3	**The answer is not reduced**: $\frac{6}{12}$ when the answer	The final step must be to check the common factors
4	**Improper fractions are not converted to mixed fractions**	Numerator > Denominator → Convert into mixed numbers (subject to the requirements of the division test)
5	**LCM error finding**(use the larger common multiple)	Short division is confirmed to be the true "minimum"
6	**Mixed fractions are added before they are converted to false**	It is recommended to use the "improper fraction conversion method" which is the least error-prone
7	**Missing sentences/units in application questions**	Must write "Answer:..." + unit, otherwise step points will be deducted